# DYUTHI DINESH

CA/NY, USA | +1-347-947-0061 | dyuthinirupama@gmail.com | Portfolio | dyuthiii.github.io | LinkedIn

## PROFESSIONAL SUMMARY

Data Analyst with **3+ years** applying advanced statistical methods, machine learning, and experimental design to drive insights from complex datasets (**1M+ records**). Published researcher specializing in **A/B testing, causal inference, and predictive modeling** across healthcare and education sectors. Proven track record optimizing LLM performance, building forecasting models, and translating data into actionable business recommendations.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages & Tools** | Python, R, SQL, PySpark, MongoDB, SPSS, Git |
| **Python Libraries** | pandas, numpy, scikit-learn, SciPy, spaCy, Seaborn, Matplotlib, MLflow, LangChain |
| **Visualization & BI** | Tableau, Power BI, ggplot2, Matplotlib, Seaborn |
| **Cloud & Platforms** | AWS (S3), Databricks, GitHub, Qualtrics, Microsoft Office Suite |
| **Statistical Analysis** | A/B Testing, Hypothesis Testing, Regression Analysis, Causal Inference, Time Series Analysis, Power Analysis |
| **Machine Learning** | Supervised/Unsupervised Learning, Predictive Modeling, NLP, Clustering (LDA, K-means), Model Evaluation, Feature Engineering |
| **Research Design** | Experimental Design (RCTs), Survey Design, IRB Compliance, Data Pipelines |
| **Domain Knowledge** | Healthcare Analytics, HIPAA Compliance, EHR Systems, Educational Analytics, Digital Health, DTC & B2B Analytics |

## EDUCATION

**Columbia University, New York, NY** — *Sept 2024 – May 2025*
Master of Arts in Quantitative Methods in the Social Sciences (QMSS) — GPA: **4.0/4.0**
*Relevant Coursework:* Machine Learning, Causal Inference, Data Mining, Natural Language Processing, Advanced Statistics

**University of British Columbia, Vancouver, BC, Canada** — *Sept 2018 – Apr 2022*
Bachelor of Science in Psychology (Honours) — GPA: **94.9%** — Graduated with Distinction
*Relevant Coursework:* Advanced Statistics, Research Methods, Experimental Design, Psychometrics, Data Analysis, Advanced Psychology/Biology & Neuroscience

## PROFESSIONAL EXPERIENCE

**Research Associate** — *June 2025 – Present*
Haupt Data Analytics and Consultancy, Remote

- Used advanced Qualtrics branching logic and respondent segmentation, enabling analysis of **100,000+ responses** for University of South Carolina ad campaign effectiveness study.
- Extracted **10+ recurring themes** from **100,000+ open-ended responses** using NLP (spaCy, sentiment analysis), reducing manual review time by **80%** and identifying key brand perception drivers for the University of South Carolina.
- Improved LLM accuracy on pharmaceutical data analysis tasks by developing a comprehensive evaluation framework benchmarking GPT-4 outputs against traditional statistical methods (t-tests, chi-squared, regression) across **500+ test cases**.
- Automated NPI database web scraping pipeline processing **10,000+ surgeon records** for LLM orthopaedic recommendation validation study, reducing manual entity matching time from **40 hours to 2 hours** using Python (BeautifulSoup, regex).
- Co-authored research manuscript comparing LLM performance to pharmacists, nurses, and practitioners, contributing **30%** of content (background, introduction, results sections) for peer-reviewed publication.

**Data Intern** — *June 2025 – August 2025*
Columbia Business School, New York, NY

- Conducted extensive data cleaning and data wrangling to create accurate datasets of student names, professor names, course names, student club names, professor evaluations and course evaluations from **20+** years of student enrollment data using Natural Language Processing methods and cut the qualitative analysis time by **90%**.
- Showed in-depth understanding of data by brainstorming ways to match columns with no clear common link between them for **10+ historical datasets**.
- Segmented **100,000+ students** into **5 clusters** using Latent Dirichlet Allocation (LDA) based on elective selection patterns across **4000+** courses, informing curriculum development, course offering decisions, and generating informal student pathways based on career interests.

- Created executive dashboards in Tableau and Power BI comparing **5 top business schools** across world rankings, used by leadership for strategic planning and competitive positioning.

**Clinical Researcher** *September 2022 – July 2024*
Wysa (AI Conversational Agent), Remote

- Analyzed multi-modal behavioral data from **1M+ users** using logistic regressions, ANOVAs, K-Means Clustering, and sentiment analysis on R (dplyr, tidyr, lubridate) and Python (pandas, scikit-learn) to generate real-world evidence of digital mental health intervention efficacy, contributing to **3 peer-reviewed publications** and regulatory submissions.
- Optimized Randomized Controlled Trial (RCT) sample sizes by **30%** through statistical power analysis and effect size calculations, reducing study costs while maintaining **80%** statistical power for intervention detection.
- Delivered **20+ high-impact analytics reports** for Fortune 500 clients (BOSCH, Aetna, UCSF, Travelers), and public-facing reports quantifying **$29M+ in potential cost savings per year** from employee mental health interventions using ROI modeling and cost-benefit analysis.

  Global Employee Mental Health Report – quantified cost savings from improved employee mental health.

  Role of AI in Mental Health Crises – identified multimodal distress patterns globally.

  Conversational AI for Mental Health – reviewed ethical and clinical risks of AI tools.

## PROJECTS

**COVID-19 Misinformation Analysis — Master's Thesis** *January 2025 – May 2025*
Columbia University & University of California San Diego

- Analyzed **800+ users'** interactions with 36 COVID-19 misinformation tweets using multiple regression to identify key drivers of COVID-19 misinformation spread, achieving **0.45 $R^2$** in explanatory model.
- Applied Silhouette Analysis/K-Means Clustering to segment **800+ users** into **3 personas** based on passive and active misinformation engagement patterns, revealing potential high-risk amplifier groups.

**LLM-Powered Mental Health Support Tool** *January 2025 – May 2025*
Columbia University Practicum

- Designed end-to-end workflow for HIPAA-compliant mental health chatbot incorporating informed consent, CBT-based interventions, and crisis escalation protocols aligned with clinical best practices.
- Established **10 safety and relevance metrics** for AI mental health systems, including response appropriateness, risk detection accuracy, therapeutic alliance measures, and clinical datasets for model training.

## PUBLICATIONS


- **Dinesh DN**, Rao MN, Sinha C. (2024). Language adaptations of mental health interventions: User interaction comparisons with an AI-enabled conversational agent (Wysa) in English and Spanish. *Digital Health*, 10. doi:10.1177/20552076241255616
- Sinha C, **Dinesh D**, Heaukulani C, Phang YS. (2024). Examining a brief web and longitudinal app-based intervention [Wysa] for mental health support in Singapore during the COVID-19 pandemic: Mixed-methods retrospective observational study. *Frontiers in Digital Health*, 6, 1443598. doi:10.3389/fdgth.2024.1443598
- Sinha C, Meheli S, **Dinesh D**, Thakkar R. (2025). Exploring the Role of App Features in Providing Continuity of Care to Users on a Digital Mental Health Platform (Wysa): A Retrospective Study of Real World Evidence. *JMIR Preprints*, 73033. doi:10.2196/preprints.73033 [*Submitted*]


## AWARDS & CERTIFICATIONS

**Awards:**
International Major Entrance Scholarship (2018-2022) — Outstanding International Student Award (2018)
Deputy Vice-Chancellor Scholarship for International Students (2019-2021) — International Student Faculty Award (2019)
The Faculty of Science International Student Award (2020) — International Undergraduate Research Award (2021)
Best Honours Presentation, PURC UBC (2022)

**Conference Presentations:**
Interdisciplinary Student Health Conference (ISHC), Canada (March 2022)
Psychology Undergraduate Research Conference (PURC), UBC (March 2022)